

4(a)	<i>artery wall</i> I <i>narrow lumen to maintain high (blood) pressure</i> I <i>ref. to valves / ref. to inner lining being wrinkled or wavy</i> 1 thick, walled / tunica media, to withstand high (blood) pressure / prevent bursting ; 2 endothelium / endothelial cells / tunica intima, are smooth, little friction to blood flow / easy flow of blood / no eddies of blood flow / AW ; 3 elastic, tissue / fibres, stretches to allow surges in blood flow / recoils to maintain blood pressure <i>or</i> force blood forward ; 4 smooth muscle (contracts to), maintains / regulates / controls blood flow ; A smooth muscle distributes blood 5 collagen fibres, avoid rupturing / bursting ;	4
4(b)(i)	red blood cells / erythrocytes ; R red and white blood cells <i>one from</i> biconcave (shape) ; no nucleus ; <i>idea of</i> uniform, cytoplasm / cell contents ; <i>idea of</i> rouleau / stacked cells ; I 'clumped' I size	2

Question	Answer	Marks
4(b)(ii)	<i>assuming arteriole unless told otherwise, accept alternative terminology for layers of wall of arteriole</i> <i>I ref. to folding</i> 1 thicker wall / more than one layer of cells in wall / has tunica intima, tunica media and tunica adventitia whereas capillary has tunica intima ; A <u>endothelium</u> for tunica intima 2 more cells forming, perimeter / tunica intima ; 3 wider (vessel) / wider lumen / AW ; A actual width(s) 4 nucleus / nuclei, present in wall only in arteriole ; 5 cells lining lumen / endothelial cells, are thicker ; 6 lumen smaller, relative to the, thickness of the wall / overall width ; 7 more (red blood) cells (in lumen) ; 8 nuclei projecting inwards only in arteriole ; 9 AVP ; e.g. <i>ref. to</i> smooth muscle cells capillaries are surrounded by cells correct calculation of actual sizes using magnifications in Fig. 4.1 arteriole 20–35 μm and capillary 6–7 μm	4

Question	Answer	Marks
4(c)(i)	1 higher (hydrostatic) pressure of blood (at start of capillary) ; 2 (pressure / ultra) <u>filtration</u> of blood ; 3 (causing) leakage / movement out / pushing out, of <u>plasma</u> ; R diffusion 4 <i>either</i> glucose / amino acids / salts, (filtered) out <i>or</i> (large) plasma proteins not (filtered) out ; 5 AVP ; e.g. <i>ref. to</i> pores / fenestrations / gaps, within / between, endothelial / lining, cells e.g. molecules smaller than MM ~68 000 (g mol⁻¹ / daltons) can pass out	2
4(c)(ii)	lymph ; A lymphatic (fluid) <i>one from</i> I <i>any cells / waste products / toxins / antibodies / fatty acids and glycerol</i> no, named / large / plasma, proteins, e.g. albumen no / little / less, oxygen / glucose higher concentration of / more, carbon dioxide AVP ; e.g. higher concentration of / more, fat / lipids / lipoproteins <i>if fluid identified as tissue fluid give one mark as an ECF for a difference between plasma and tissue fluid</i> e.g. no, named / large / plasma, proteins e.g. albumen	2